- **Engineering preparation and ground improvement works to make the site suitable for the installation of the concrete platform.**
- **Excavation of cable trenches in accordance with the project design for the installation of DC, LV, and AC cabling.**
- **Preparation of the structural design and structural calculations for the concrete platform.**
- **Commencement of the concrete platform construction.**
- **Lay a weed-suppressing geotextile the gravel surrounding the platform to prevent vegetation growth.**

- **Civil Works**

Site establishment and mobilization.
Site survey and setting out.
Earthworks, grading, and site levelling.
Ground compaction and geotechnical preparation.
Excavation works for foundations and cable trenches.
Installation of underground conduits and cable ducts.
Installation of drainage system and stormwater management.
Construction of reinforced concrete foundations/platforms.
Concrete curing and quality inspections.
Installation of anchor bolts and embedded steel plates.
Backfilling and reinstatement works.

- **Electrical Infrastructure**

Installation of DC, LV, MV, and communication cable trenches.
Installation of cable trays and underground duct banks.
Pulling and termination of DC, LV, MV, and fibre optic cables.
Earthing (grounding) system installation.
Lightning protection system installation.
Installation of auxiliary power supply.
Installation of site lighting.

- **BESS Installation**

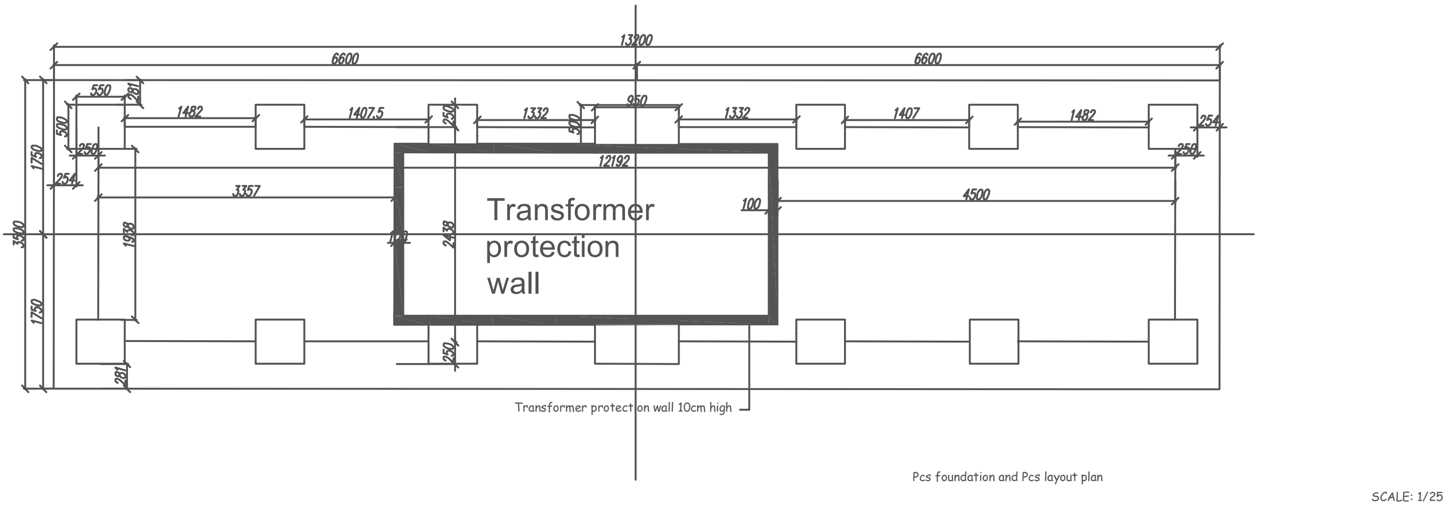
Delivery and offloading of BESS containers/cabinets.
Positioning and alignment of BESS units.
Mechanical installation of battery cabinets.
Installation of PCS (Power Conversion System).
Installation of MV transformer (if applicable).
Installation of RMU/Switchgear.
Installation of HVAC system.
Installation of fire detection and fire suppression systems.
Installation of EMS/SCADA communication equipment.
Installation of monitoring and control panels.

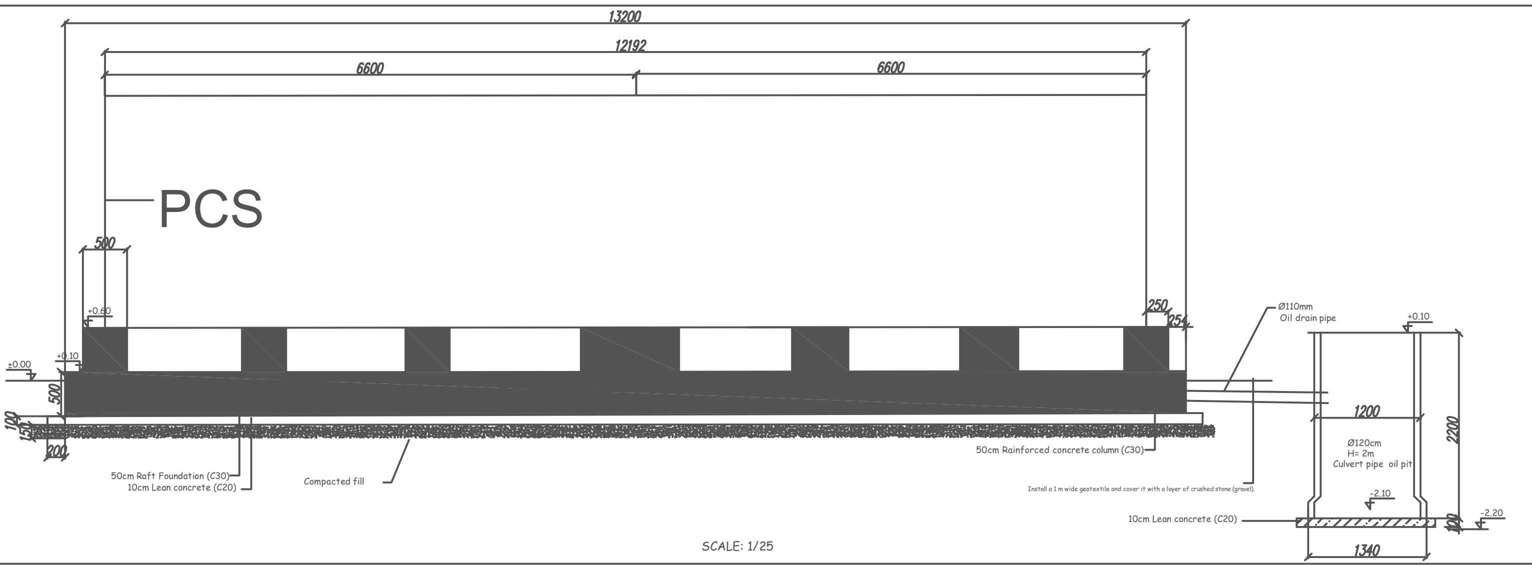
- **Testing & Commissioning**

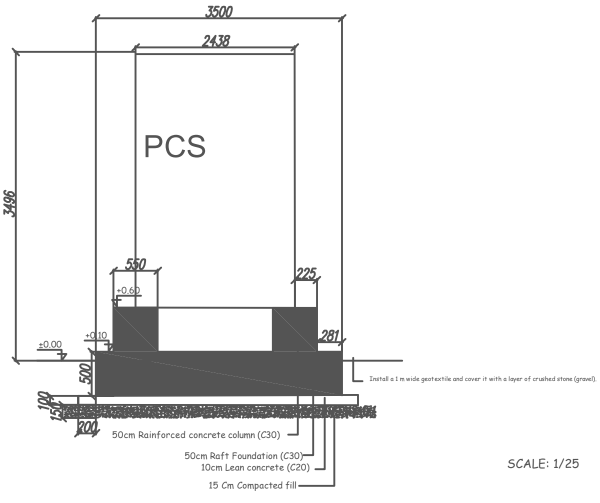
Mechanical completion inspection.
Cable insulation resistance testing.
Continuity and earth resistance testing.
Functional testing of protection systems.
SCADA communication testing.
Battery commissioning.
PCS commissioning.
Grid synchronization tests.
Performance and acceptance testing.
Client training.
Final documentation and handover.

- **Site Completion**

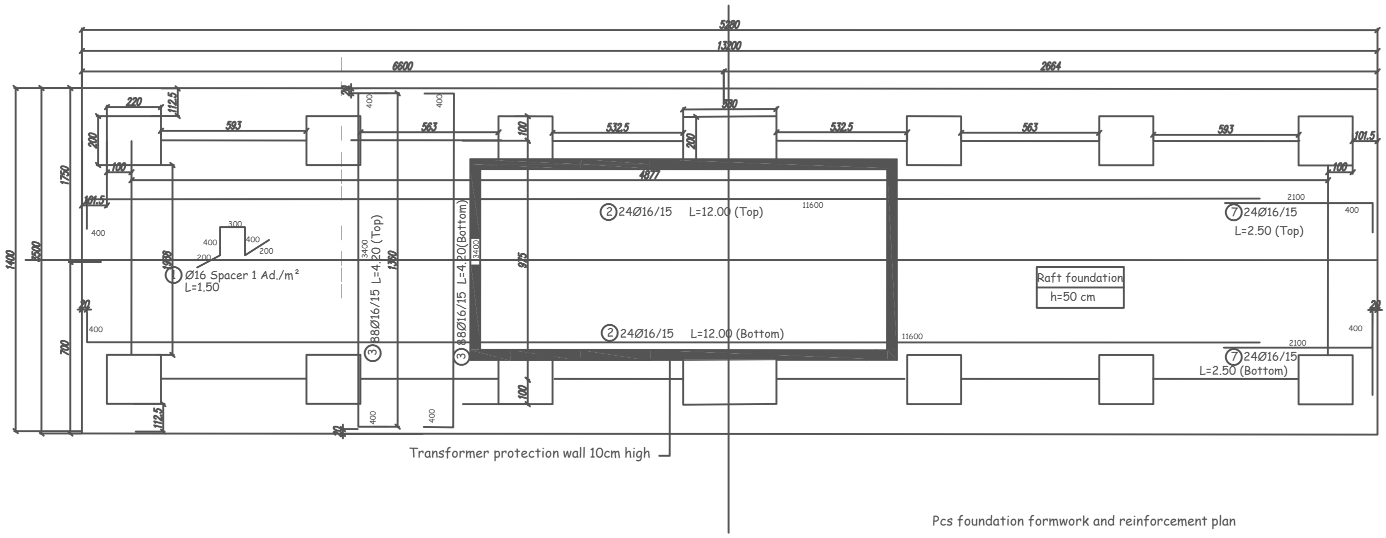
Site reinstatement and clean-up.
As-built documentation.
Final inspection.
Punch list rectification.
Taking Over Certificate (TOC).
Project handover.





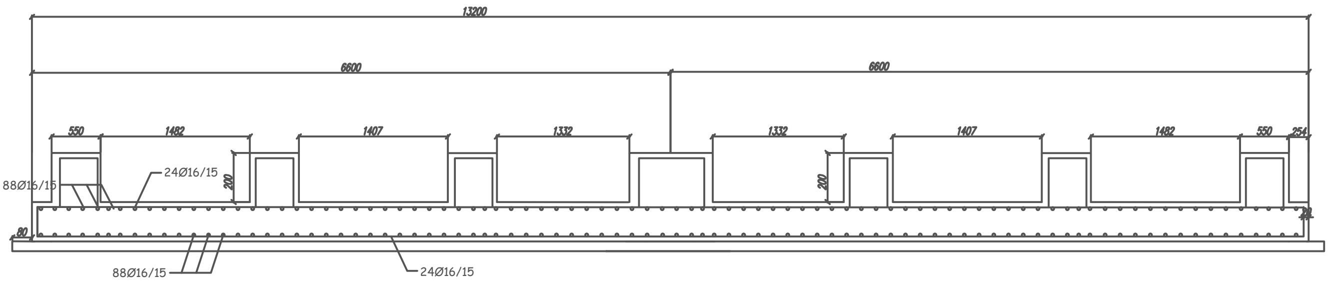


NOTES : At ground level, a 1 m wide geotextile shall be laid around the entire perimeter of the platform and covered with a layer of gravel.



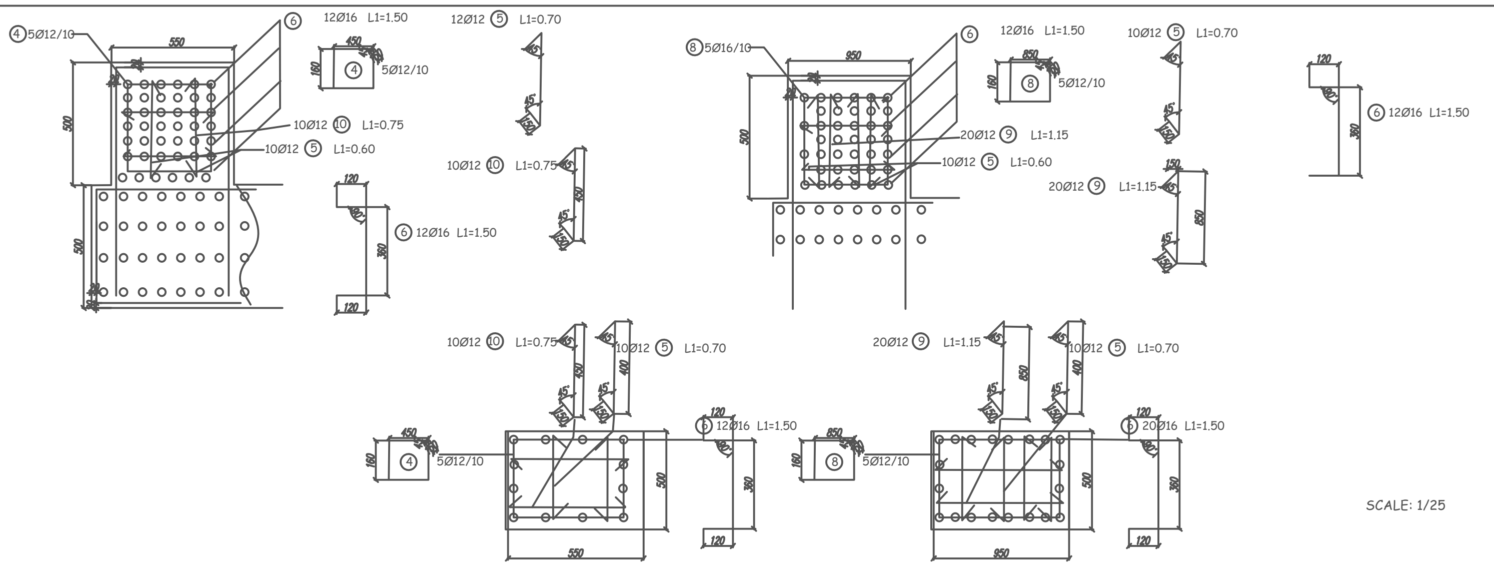
Pcs foundation formwork and reinforcement plan

SCALE: 1/25



PCS foundation

SCALE: 1/25



SCALE: 1/25

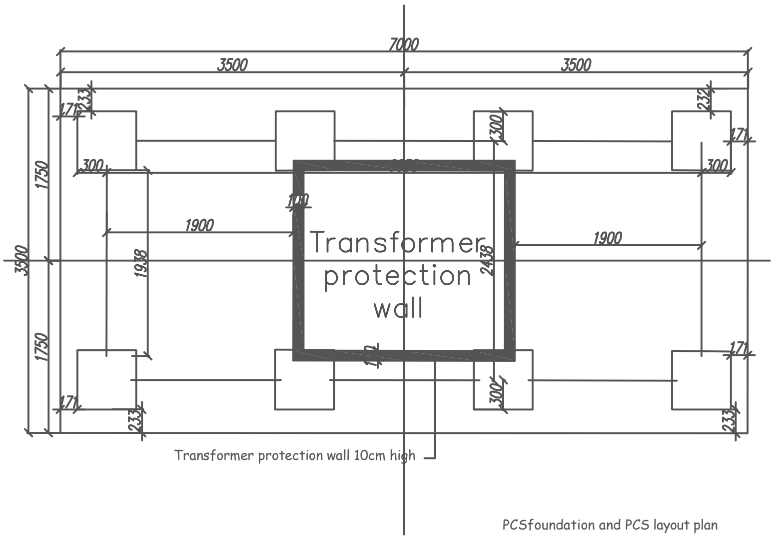
FOUNDATION REBAR CONCRETE AND FORMWORK QUANTITIES						
Pos	Diameter	Shape	Quant.	Length(m)	Total Length	Weight(kg)
1	Ø16	200 400 300 200	48	1.50	72	114
2	Ø16	400 11600	48	12.00	576	910
3	Ø16	400 3400 400	176	4.20	740	1.169
4	Ø12	150 400 450	60	1.90	114	102
5	Ø12	150 400 150	200	0.70	140	125
6	Ø16	300 900 300	184	1.50	276	436
7	Ø16	2100 400	48	2.50	120	190
8	Ø12	150 400 850	10	2.80	28	25
9	Ø12	150 850 150	20	1.15	23	21
10	Ø12	150 450 150	120	0.75	90	80
					Total rebar	3172
					Total Concrete	28 m³

MATERIAL

CONCRETE GRADE: C35/45 (fck: 35MPa)

STEEL GRADE : B500C (fyk: 500MPa)

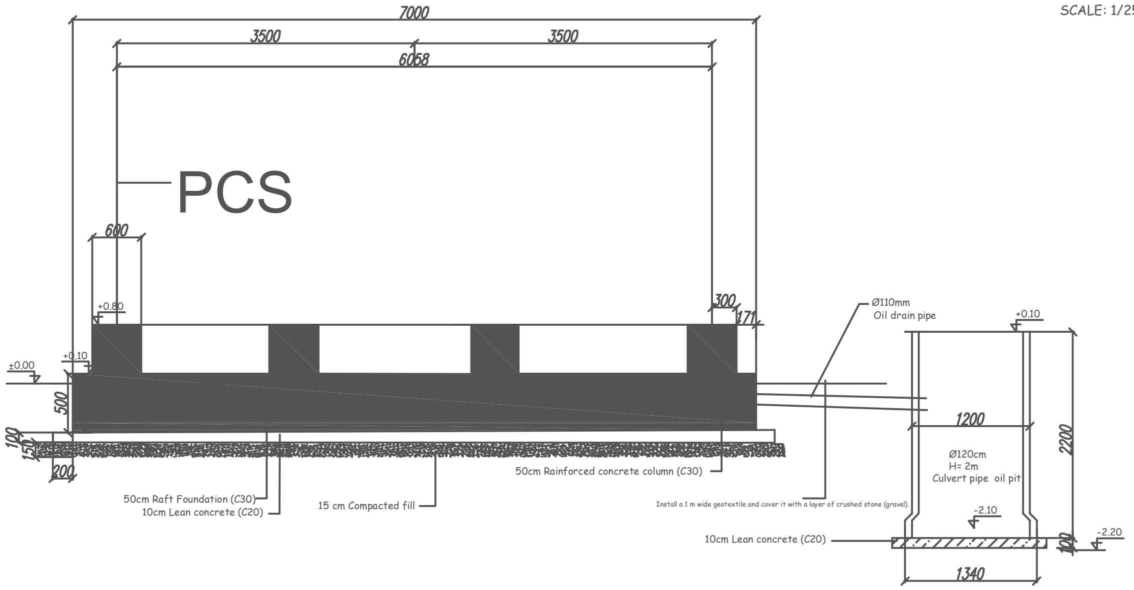
AgR	S	G.T.	I.C	γl	D.C.	SPECT.	q	TC	Con c.
0.25	1.15	C	III	1.2		TYPE 1	3.0	[illegible]	5.0cm

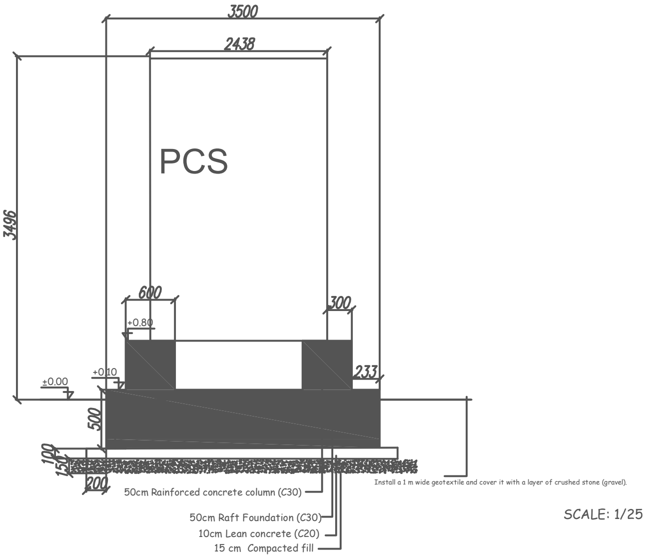


PCSfoundation and PCS layout plan

SCALE: 1/25

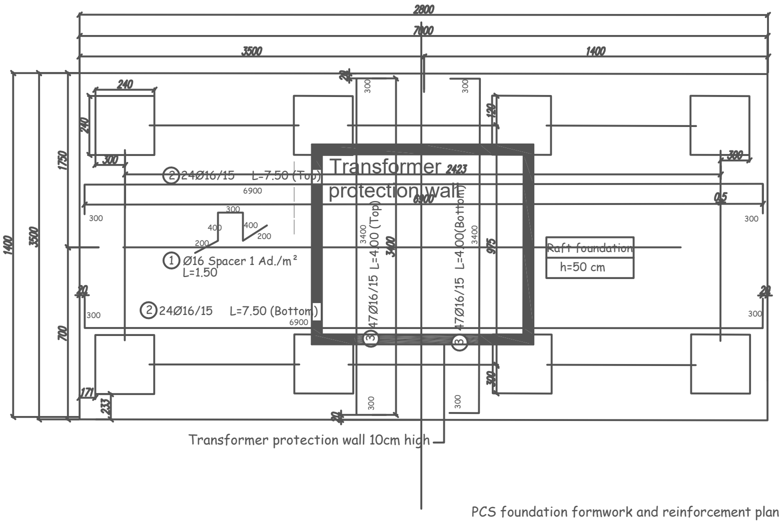
SCALE: 1/25





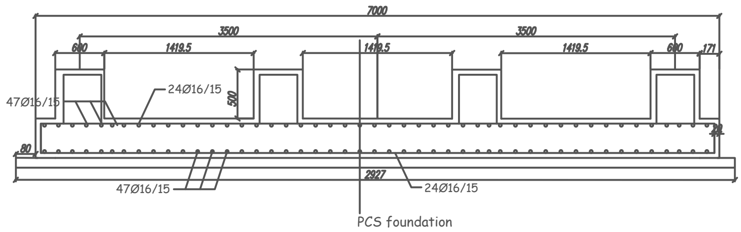
SCALE: 1/25

NOTES : At ground level, a 1 m wide geotextile shall be laid around the entire perimeter of the platform and covered with a layer of gravel.



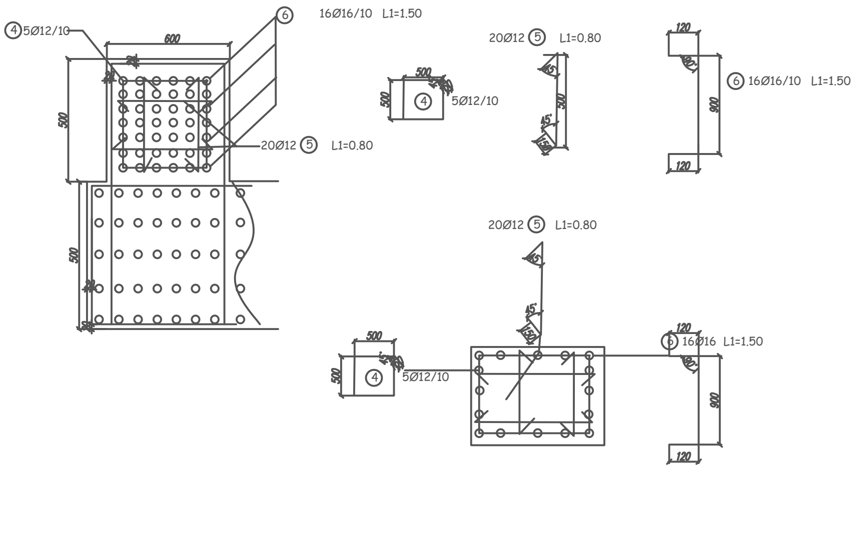
PCS foundation formwork and reinforcement plan

SCALE: 1/25



PCS foundation

SCALE: 1/25



SCALE: 1/25

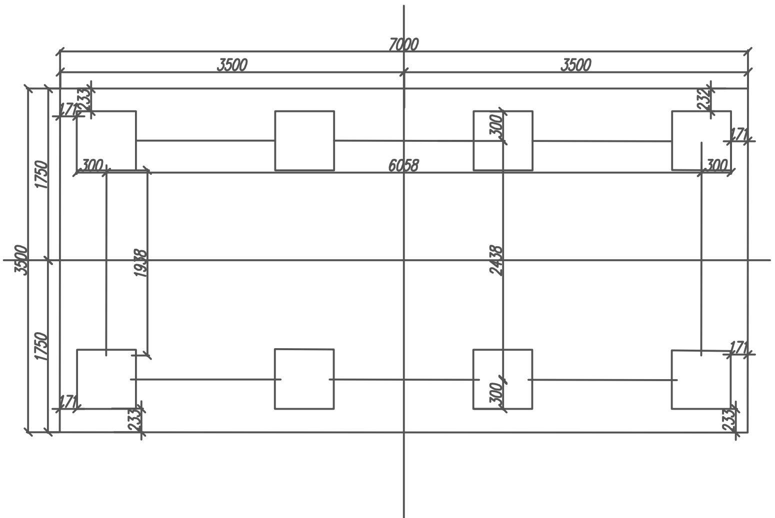
FOUNDATION REBAR CONCRETE AND FORMWORK QUANTITIES						
Pos	Diameter	Shape	Quant.	Length(m)	Total Length	Weight(kg)
1	Ø16	200 400 300	27	1.50	41	65
2	Ø16	300 6900 300	48	7.50	360	569
3	Ø16	300 3400 300	94	4.00	376	594
4	Ø12	150 500 500	40	2.30	92	82
5	Ø12	150 500 150	160	0.80	128	114
6	Ø16	300 900 300	128	1.50	192	304
					Total rebar	1728
					Total Concrete	15 m³

MATERIAL

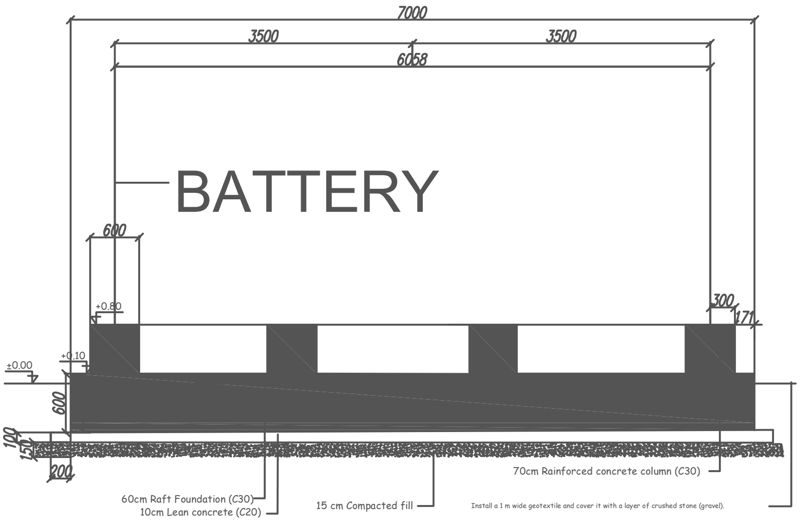
CONCRETE GRADE: C35/45 (fck: 35MPa)

STEEL GRADE : B500C (fyk: 500MPa)

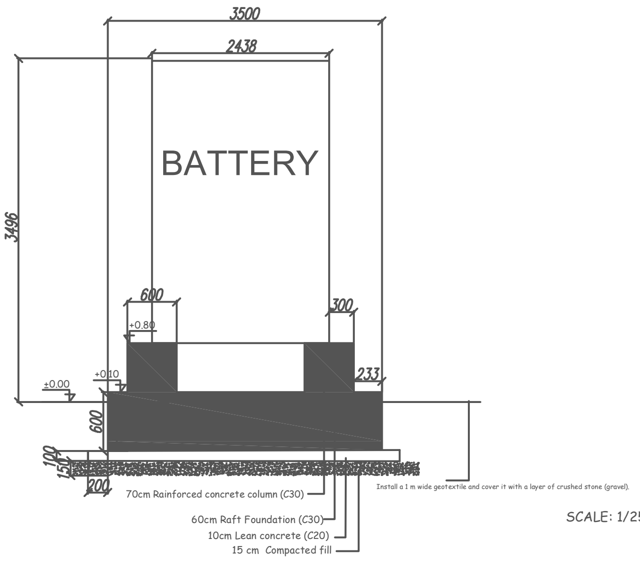
AgR	S	G.T.	I.C	γI	D.C.	SPECT	q	TC	Con c.
0.25	1.15	C	III	1.2		TYPE 1	3.0	[illegible]	5.0cm



Batteryfoundation and battery layout plan
SCALE: 1/25

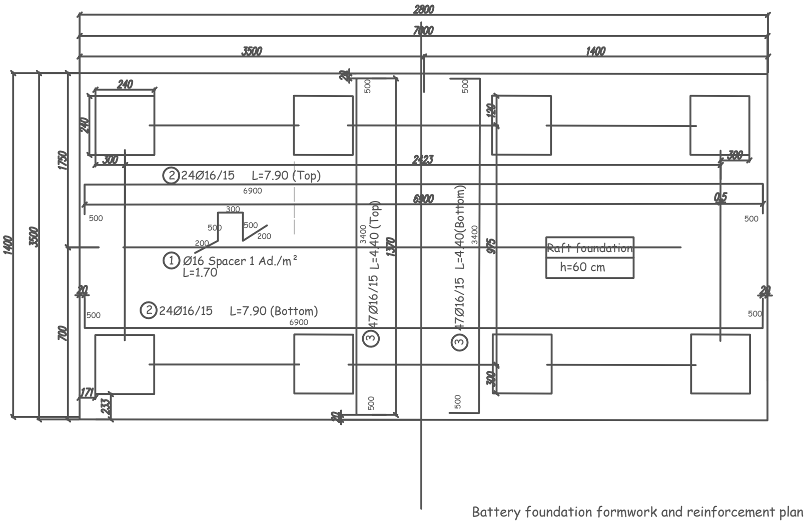


SCALE: 1/25



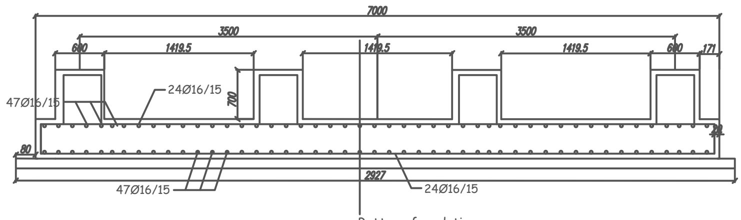
SCALE: 1/25

NOTES : At ground level, a 1 m wide geotextile shall be laid around the entire perimeter of the platform and covered with a layer of gravel.



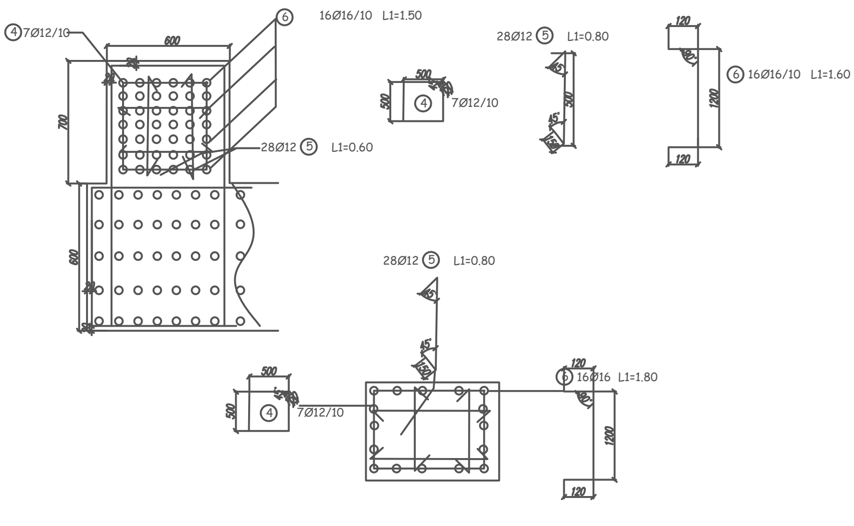
Battery foundation formwork and reinforcement plan

SCALE: 1/25



Battery foundation

SCALE: 1/25



SCALE: 1/25

FOUNDATION REBAR CONCRETE AND FORMWORK QUANTITIES						
Pos	Diameter	Shape	Quant.	Length(m)	Total Length	Weight(kg)
1	Ø16	200 500 300	26	1.70	45	71
2	Ø16	500 6900 500	48	7.90	380	600
3	Ø16	500 3400 500	94	4.40	414	654
4	Ø12	150 500 500	56	2.30	129	115
5	Ø12	150 500 150	224	0.80	180	160
6	Ø16	300 1200 300	128	1.80	231	365
					Total rebar	1965
					Total Concrete	17 m³

MATERIAL

CONCRETE GRADE: C35/45 (fck: 35MPa)

STEEL GRADE : B500C (fyk: 500MPa)

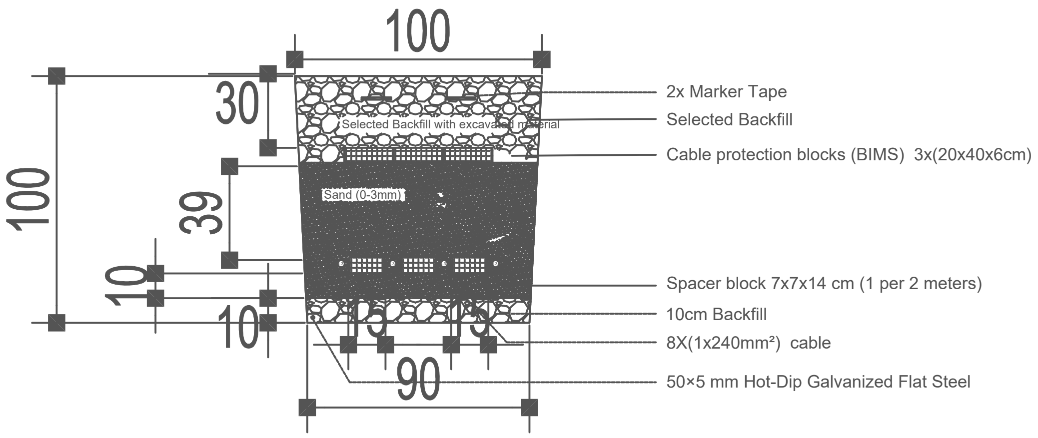
AgR	S	G.T.	I.C	γl	D.C.	SPECT.	q	TC	Con c.
0.25	1.15	C	III	1.2		TYPE 1	3.0	[illegible]	5.0cm

	MESAURE (M)	CABLE	CABLE TYPE	Connecting Terminals for Cables
DC	1160	1×240 mm² , 1.5 kV DC, BESS Battery-to-PCS, OEM Approved	H1Z2Z2-K	160
MV	120	1 X 240 mm² 12/20 (24) kV,screened XLPE	N2XSH/N2XSY	12
AC	120	4×35 mm² + 1×16 mm² PE 0.6/1kv	N2XH	20
COM (RS485/CAN+ FSS	250	2 X 2 X 0.75 mm² Shielded Twis tead Pair Communication Cable,120Ω, suitable	LI2YCY-TP	
Ethernet COM:	200	CAT6 FTP Industrial	FTP	
EARTHING	70	1×150 mm² Cu G/Y Earthing Conductor	Outdoor/protected installation suitable type – TBD	20
EARTHING	150	50×5 mm Hot-Dip Galvanized Flat Steel		Cross / T / Parallel Earthing Clamp, suitable for 50×5 mm hot-dip galvanized flat steel 20
				Cu-to-HDG Steel Earthing Connector / Bimetallic Earthing Connector, suitable for 150 mm² Cu cable to 50×5 mm hot-dip galvanized flat steel. 16

Section 1-1

90x100cm

DC+GROUND



Section 1-2

90x100cm

MV+GROUND